

End-to-End Process · Procure-to-Pay (P2P)

PCF 4.2 Develop Sourcing Strategy · PCF 4.2 Procure Materials and Services · PCF 4.3 Manage Logistics · PCF 9.2 Process Accounts Payable

View 1 — HITL Design

P2P HITL Design Summary

P2P Aggregate H% · M3	Decision Gates	Transaction Steps	Exception Flows	Δ M1 → M3
55% Human	5 · 80% H	13 · 35% H	4 · 82% H	-17%

H% by Pattern Type at Each Maturity Level

Pattern	Nature of Work	M1 Undesigned	M2 Emerging	M3 ★ Designed	M4 Optimized	M5 Leading
Decision	Judgment & Authority	92%	88%	80%	72%	65%
Knowledge	Synthesis & Interpretation	85%	78%	65%	52%	42%
Document	Content Generation	72%	58%	40%	28%	18%
Transaction	Rules-Based Processing	60%	48%	35%	22%	12%
Exception	Non-Standard Resolution	95%	92%	82%	75%	68%

Pattern Type Legend

Decision 80% H at M3	Knowledge 65% H	Document 40% H	Transaction 35% H	Exception 82% H
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P2P End-to-End Process — HITL Design · PCF-Derived Process Ownership

Swim lanes = PCF process group ownership · PCF L3 reference shown on each step · M3 H% = HITL design intent

Materials PCF 4.2–4.3 PCF 4.2–4.3	Purchasing PCF 4.2 PCF 4.2	Finance PCF 9.2 PCF 9.2	Supplier External External participant
4.2.4 Process Requisition DOCUMENT 40%	4.2.1 Develop Sourcing Strategy KNOWLEDGE 65%	9.2.1 Validate Invoice TRANSACTION 35%	Ext Receive Purchase Order TRANSACTION 35%
Decision gate 4.2.4 Approve Requisition DECISION 80%	4.2.1 Develop Category Plan KNOWLEDGE 65%	Decision gate 9.2.3 Approve for Payment DECISION 80%	Ext Ship Goods / Deliver TRANSACTION 35%
4.2.4 → Purchasing (4.2.4)	Decision gate 4.2.3 Identify & Select Suppliers DECISION 80%	9.2.3 Process Payment TRANSACTION 35%	Ext → Materials (4.3.2)
4.3.2 Receive / Inspect / Store Product TRANSACTION 35%	4.2.3 → Procure Materials (4.2.4)	9.2.3 → Supplier (Receive Payment)	Ext Prepare Invoice DOCUMENT 40%
4.3.2 → Finance (9.2.1)			

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	4.2.4 Solicit Supplier Quotes DOCUMENT 40%	9.2.4 Reconcile Accounts TRANSACTION 35%	Ext Issue Invoice TRANSACTION 35%
	4.2.4 Create Purchase Order DOCUMENT 40%		Ext → Finance (9.2.1)
	4.2.4 Distribute Purchase Order TRANSACTION 35%		Ext Receive Payment TRANSACTION 35%
	4.2.4 → Finance (9.2.1) + Supplier		
Exception Flows — Human Resolution Required (82% H at M3)			
4.3.2 Resolve Receipt Exception EXCEPTION 82%	4.2.4 Resolve Order Exception EXCEPTION 82%	9.2.2 Resolve Invoice Discrepancies EXCEPTION 82%	
		9.2.1 Invoice Exception EXCEPTION 82%	

HITL Design Insight

HITL Design Insight — P2P

5 Decision gates (80% H at M3) are the irreducibly human layer — supplier selection, requisition approval, and payment authorization carry strategic and financial authority by design. **13 Transaction steps** (35% H) are the AI absorption opportunity. **4 Exception flows** (82% H) require human resolution. P2P aggregate at M3: **55% Human / 45% AI** — grounded in APQC PCF SIPOC reference architecture.

HITL Impact Analysis

H% = Human involvement at each maturity level · M3 = HITL Design Intent

Process Step (L3)	PCF	Pattern	M1 H%	M3 H% ★	Δ M1→M3	AI Opportunity
P2P End-to-End Process · L1 Aggregate	72%		72%	55%	-17%	Significant
PCF 4.2 · Develop Sourcing Strategy						
↳ Develop Sourcing Strategy	4.2.1	Knowledge	82%	65%	-17%	High
↳ Develop Category Plan	4.2.1	Knowledge	82%	65%	-17%	High
↳ Identify & Select Suppliers	4.2.3	Decision	92%	80%	-12%	Retain Human
PCF 4.2 · Procure Materials and Services						
↳ Process Requisition	4.2.4	Document	72%	40%	-32%	High
↳ Approve Requisition	4.2.4	Decision	92%	80%	-12%	Retain Human
↳ Solicit Supplier Quotes	4.2.4	Document	72%	40%	-32%	High
↳ Create Purchase Order	4.2.4	Document	72%	40%	-32%	High

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↳ Distribute Purchase Order	4.2.4	Transaction	60%	35%	-25%	High
↳ Resolve Order Exception	4.2.4	Exception	95%	82%	-13%	Retain Human
PCF 4.3 · Manage Logistics and Warehousing						
↳ Receive / Inspect / Store Product	4.3.2	Transaction	60%	35%	-25%	High
↳ Resolve Receipt Exception	4.3.2	Exception	95%	82%	-13%	Retain Human
PCF 9.2 · Process Accounts Payable						
↳ Validate Invoice	9.2.1	Transaction	60%	35%	-25%	High
↳ Resolve Invoice Discrepancies	9.2.2	Exception	95%	82%	-13%	Retain Human
↳ Approve for Payment	9.2.3	Decision	92%	80%	-12%	Retain Human
↳ Process Payment	9.2.3	Transaction	60%	35%	-25%	High
↳ Reconcile Accounts	9.2.4	Transaction	60%	35%	-25%	High
↳ Invoice Exception	9.2.1	Exception	95%	82%	-13%	Retain Human
External Participant · Supplier						
↳ Receive Purchase Order	Ext	Transaction	60%	35%	-25%	High
↳ Ship Goods / Deliver	Ext	Transaction	60%	35%	-25%	High
↳ Prepare Invoice	Ext	Document	72%	40%	-32%	High
↳ Issue Invoice	Ext	Transaction	60%	35%	-25%	High
↳ Receive Payment	Ext	Transaction	60%	35%	-25%	High

AI Capability Map — P2P

PCF 4.2 Sourcing & Procurement	PCF 4.3 Logistics & Warehousing	PCF 9.2 Accounts Payable	External Supplier
• Spend analytics & opportunity identification	• PO-to-receipt 3-way matching	• Invoice OCR & data extraction	• EDI / portal PO transmission
• Supplier scoring & risk ranking	• Goods receipt anomaly detection	• 2-way / 3-way match automation	• ASN generation & tracking
• RFQ generation & quote analysis	• Barcode / RFID inventory tracking	• Exception triage & root cause analysis	• AI-assisted invoice drafting
• PO auto-generation from contract terms	• Discrepancy classification & routing	• Payment run automation	• Remittance auto-match & clearing
• Requisition pre-population from history	• Carrier performance monitoring	• AP ledger reconciliation	• Supplier self-service portal
• Policy & budget compliance validation	• Inventory replenishment signals	• Duplicate invoice detection	• Payment confirmation automation